

Agriculture Monitoring System Using Graph Data Structure in C

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Abstract

Agriculture monitoring plays a vital role in improving farm productivity and resource management. This project presents a console-based Agriculture Monitoring System implemented in C programming using a Graph data structure with adjacency list representation. Each farm is modeled as a vertex, and connections between farms are represented as directed edges. The system supports farm management operations such as addition, deletion, update, search, and connection handling. Dynamic memory allocation is used for efficient memory management. The project demonstrates practical implementation of graph concepts in real-world agricultural monitoring scenarios.

Keywords

Agriculture Monitoring, Graph Data Structure, Adjacency List, C Programming, Dynamic Memory Allocation, Linked List

I. Introduction

Agriculture monitoring systems help track environmental parameters such as temperature and moisture to improve crop yield and optimize farm management. Efficient data handling is required when managing multiple farms and their interconnections.

Graph data structures are widely used to represent relationships between entities. In this project, each farm is treated as a node (vertex), and connections between farms are represented as edges using adjacency lists.

This implementation demonstrates how graph structures can be applied in real-world agricultural systems using C programming.

II. Problem Statement

Traditional farm management systems lack structured representation of relationships between farms. There is a need for:

- Efficient storage of farm data
- Connection mapping between farms
- Dynamic memory handling
- Easy search and update operations

This project aims to design a graph-based agriculture monitoring system to address these requirements.

III. System Design

A. Data Structures Used

1. Graph
2. Adjacency List
3. Linked List
4. Structures
5. Dynamic Memory Allocation

B. Farm Structure

Each farm contains:

- Farm Name
- Temperature
- Moisture Level
- Pointer to adjacency list

C. Edge Structure

Each edge contains:

- Destination Farm Index
 - Pointer to next edge
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IV. Methodology

1. Create an empty graph.
 2. Dynamically allocate memory for farms.
 3. Store farm details (Name, Temperature, Moisture).
 4. Add connections using adjacency list.
 5. Provide menu-driven operations:
 - Add Farm
 - Update Farm
 - Delete Farm
 - Search Farm
 - Display Farms
 - Add/Remove Connections
 - Count Farms
 6. Free allocated memory before program termination.
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V. Implementation Details

The system is implemented in C using:

- `malloc()` for dynamic memory allocation
- `realloc()` for resizing farm array
- `free()` for memory deallocation
- `strcmp()` for searching farm names

The graph is implemented using adjacency list representation for efficient memory usage.

VI. Results and Output

The system successfully:

- Adds farms dynamically
- Maintains directed connections
- Displays adjacency list
- Updates environmental data
- Frees memory properly

Example Output:

FarmA -> FarmB
FarmB ->

VII. Advantages

- Efficient memory usage
 - Dynamic scalability
 - Easy data modification
 - Graph-based relational representation
 - Educational implementation of data structures
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VIII. Limitations

- Console-based interface only

- No file storage (data lost after exit)
 - Directed connections only
 - No graphical visualization
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IX. Future Enhancements

- File handling for permanent storage
 - BFS and DFS traversal
 - Alert system for extreme temperature/moisture
 - GUI-based version
 - IoT sensor integration
 - Web-based monitoring dashboard
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X. Conclusion

The Agriculture Monitoring System demonstrates practical implementation of graph data structures in C programming. By modeling farms as vertices and their connections as edges, the system efficiently manages agricultural data and relationships. The project enhances understanding of adjacency lists, dynamic memory management, and structured programming techniques.

References

1. Weiss, M. A., *Data Structures and Algorithm Analysis in C*, Pearson Education.
2. Kernighan, B. W., and Ritchie, D. M., *The C Programming Language*, Prentice Hall.
3. Cormen, T. H., et al., *Introduction to Algorithms*, MIT Press.